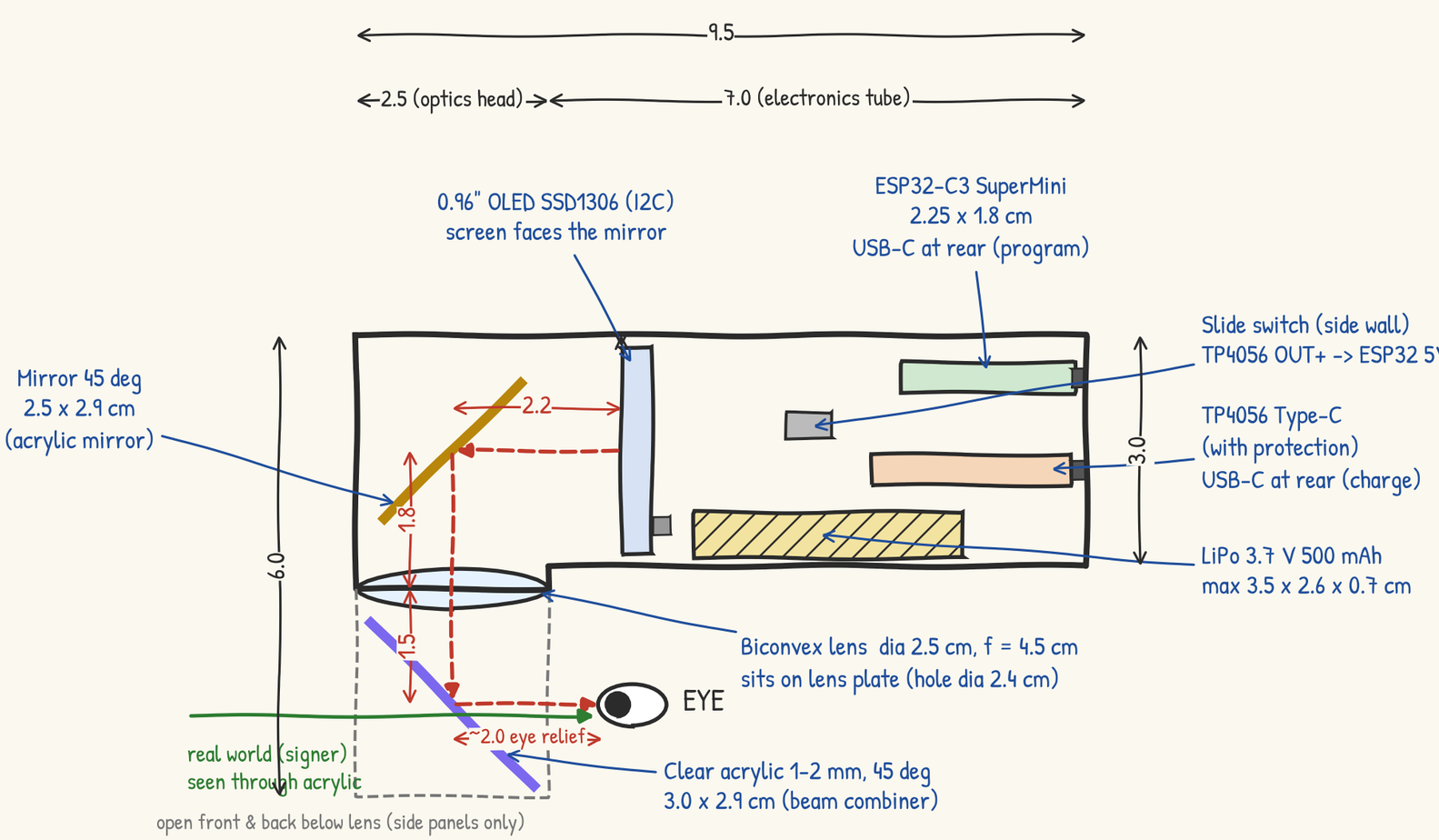
Sign-Language HUD Glasses - Display Unit (Side View, to scale)

Sheet 1 of 2 | All dimensions in cm (internal) | Based on June_Glasses layout, adapted for ESP32-C3 + 25 mm / f = 45 mm biconvex lens



SIDE VIEW (left = front, looking at right eye from outside)
width into page = 2.9 cm internal

Why these distances?

OLED -> mirror = 2.2 cm
mirror -> lens = 1.8 cm
object distance u = 2.2 + 1.8 = 4.0 cm
lens focal length f = 4.5 cm (u < f)

1/v = 1/f - 1/u = 1/4.5 - 1/4.0
v = -36 cm (virtual, upright image)
magnification = |v/u| = 9x

=> text appears ~38 cm in front of the eye,
large and easy to focus

Build notes

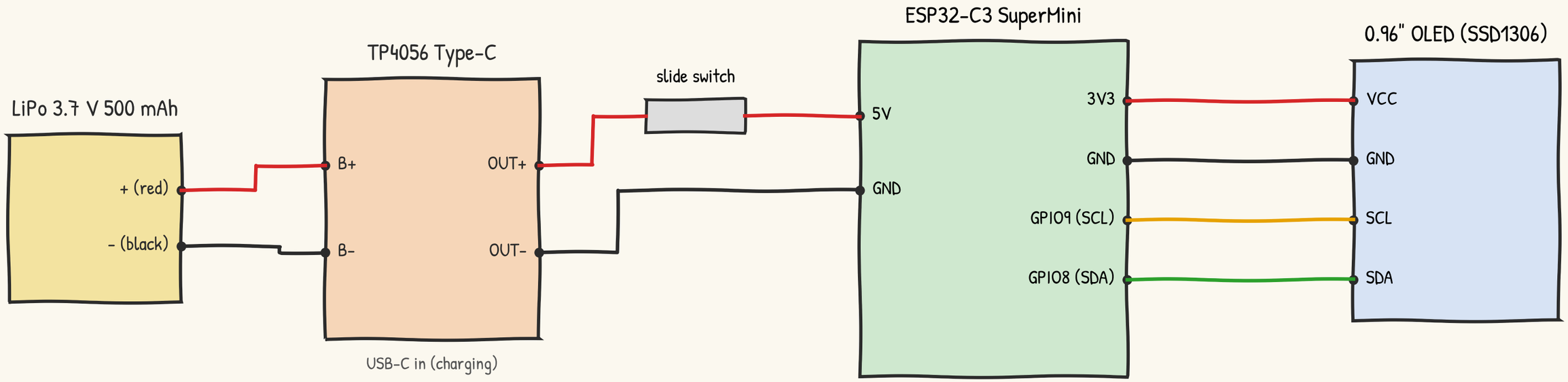
1. Focus tuning: slide OLED +/- 3 mm until text is sharp (keep OLED->lens path < 4.5 cm).
2. Align OLED active-area centre with mirror centre.
3. Paint interior matte black (stops stray glare).
4. Use thin acrylic (1-2 mm) to avoid double image.
5. Two 45 deg reflections: if text looks flipped, fix in code: display.setRotation(2) / mirror cmd.
6. Material: 3 mm foam board / cardboard (proto), then 3D-printed PLA.
7. Unit clips on glasses arm, head in front of eye.
8. Option: move LiPo to opposite arm for balance.

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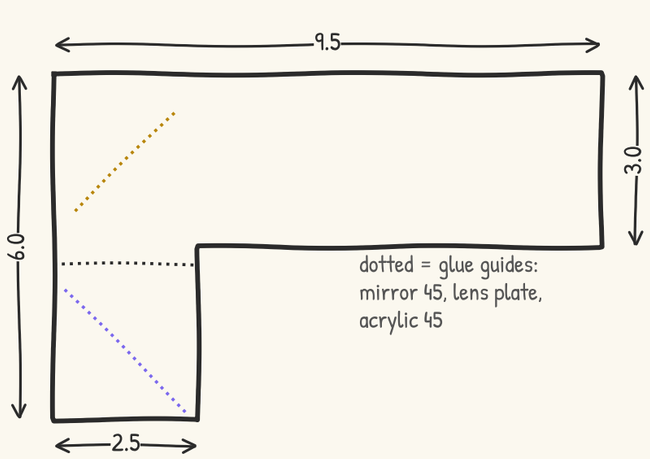
Sign-Language HUD Glasses – Wiring & Cut Templates

Sheet 2 of 2 | Cut sizes in cm | 3 mm foam board / cardboard

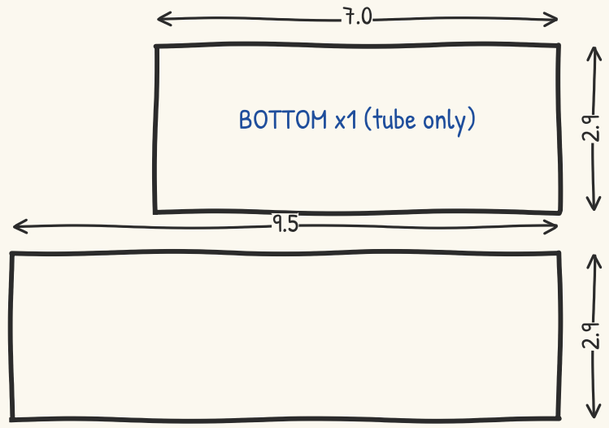
WIRING DIAGRAM



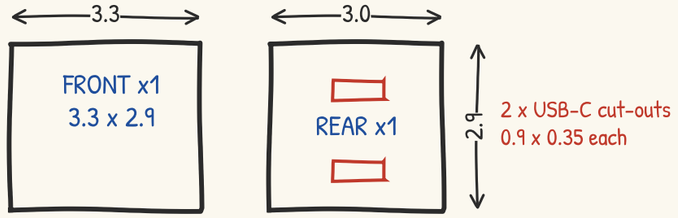
Power: LiPo -> TP4056 (charge + protection) -> switch -> ESP32-C3 5V pin (on-board 3.3 V regulator) | OLED on I2C, address 0x3C | red = +, black = GND, yellow = SCL, green = SDA

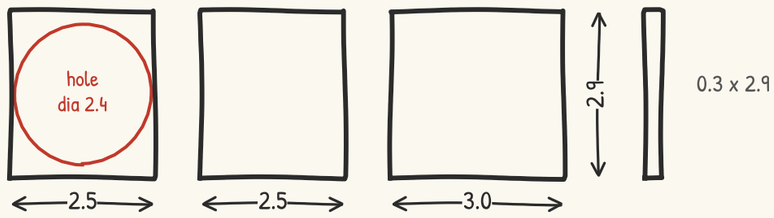


SIDE PANEL (L & R)
x2



TOP (full length)
x1





All panels 2.9 cm wide = internal width. Glue with hot glue; tape seams black.

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